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INSTRUCTIONS TO CANDIDATES

1. All examination rules stated by the Tshwane University of Technology apply.
2. Ensure a single final version of your source code is handed in as requested.
3. State all necessary assumptions clearly in code commentary where needed.

MARKS: 50

PAGES: 11 (incl. cover)

EXAMINER:

Mr A.J. Smith

Mr M. Chauke

Prof J.A. Jordaan

MODERATOR:

Mr D Engelbrecht

TIME:

90 Minutes

15 Minutes extra time in the event of computer problems

FACULTY OF ENGINEERING AND THE BUILT ENVIRONMENT

DEPARTMENT OF ELECTRICAL ENGINEERING

ES216BB ENGINEERING SOFTWARE DESIGN B

EVALUATION 1 B

AUGUST 2026

EVALUATION INSTRUCTIONS

1. **Plagiarism:** Submit only original work. Similarity software will verify the authenticity of all submissions.

2. **Permitted Tools:** Use only CodeBlocks and Google Chrome to access the evaluation, view the paper, and upload your submission. Email, other online resources, memory sticks and unauthorised tools are prohibited.

3. **File Submission:** Rename the supplied source template to <student number>.cpp. Do not add a name, surname or other text to the file name.

4. **Uploading Instructions:** ONLY SUBMIT YOUR SOURCE FILE (.cpp) through the designated upload link. The latest submission is retained.

5. **Evaluation Scope:** This assessment covers Engineering Software Design B Units 1 to 3.

6. **Programming Language:** Construct the required program in C++11 and follow structured programming principles.

7. **Editing and Requirements:** Meet all specified requirements and use the attached appendices as required.

8. **Evaluation Requirements:**

    a. Remember to save your work on the PC D: drive and save regularly throughout the evaluation.
    b. Do not modify the given libraries, type declarations, prototypes, menus or comments in the supplied .cpp template.
    c. Complete every required function in the comment blocks and use the exact names and parameters supplied in the question paper and template.
    d. The supplied input file is **turbine_service.txt.**

C++ FILE CODE EXPLANATION

The supplied C++ file sets up a menu-driven Wind Turbine Maintenance Planner. It imports records from turbine_service.txt, stores them in a dynamically allocated array, and exports the required report to turbine_maintenance_plan.txt.

The system includes functionalities for:

- **Load:** Turbine service records from a text file.
- **Calculate:** A service-priority classification.
- **Display:** Due and corrective maintenance.
- **Export:** A maintenance plan.

The program uses a typedef structure named ServiceTask with direct data members, a static reference record accessed with the dot operator, and a dynamic array accessed through a pointer and arrow operator.

The required type declarations and functions below must be implemented in the appropriate comment blocks in the supplied .cpp template file.
Use the exact function names and parameters from the supplied template.

STRUCTURE DEFINITION EXPLANATION

The ServiceTask structure and supporting types define the properties and operations required by this engineering system:

- **Structure and type requirements:**
  - **turbineID** (int): unique turbine identifier.
  - **daysSinceService** (int) and operatingHours (float).
  - **typeCode** (char): R for routine or F for corrective.
  - **serviceValue** (float): direct labour-hours or fault value.
  - **tasks** pointer and arraySize: dynamic maintenance data.

- **Required functions:**
  - InitialiseServiceTask(): initialise one record.
  - TextFileLineCount(): validate and count file records.
  - ReadFileAndPopulate(): create/reload dynamic data.
  - CalculateServicePriority(): perform the scenario calculation.
  - DisplayServiceTask(): display scenario information.
  - DisplayServiceTasks(): complete the rotated Unit 1 technique.
  - Report and cleanup functions: export results and release dynamic memory.

FUNCTION EXPLANATIONS

1. InitialiseServiceTask Function

void ServiceTask::InitialiseServiceTask(int, int, float, char, float)

Purpose: Initialises a service task using direct structure members.

Parameters:
identifier, days, hours, type code and value.

Returns:
No value.

---

2. TextFileLineCount Function

int TextFileLineCount(string fileName)

Purpose: Checks the supplied file and counts non-empty service records.

Parameters:
fileName: input text-file name.

Returns:
Number of valid records, or zero.

---

3. ReadFileAndPopulate Function

void ReadFileAndPopulate(string, ServiceTask **, int *)

Purpose: Replaces prior data with a dynamically allocated, arrow-populated task array.

Parameters:
file name, task-array pointer and size pointer.

Returns:
No value; updates the caller data.

---

4. CalculateServicePriority Function

string CalculateServicePriority(ServiceTask task)

Purpose: Returns SCHEDULED, DUE or CRITICAL from the service values and type.

Parameters:
task: one service-task structure.

Returns:
Service priority text.

---

5. DisplayServiceTask Function

void DisplayServiceTask(ServiceTask task)

Purpose: Overload 1 displays one task with priority and days-since-service data.

Parameters: task: one service-task structure.

Returns: No value; outputs to the console.

6. DisplayServiceTasks Function

void DisplayServiceTasks(ServiceTask *, int, int = 180)

Purpose: Overload 2 displays due/corrective work using the default due-day threshold.

Parameters: array pointer, size and optional due-day threshold.

Returns: No value; outputs to the console.

7. WriteMaintenancePlan Function

void WriteMaintenancePlan(ServiceTask *, int)

Purpose: Exports non-scheduled maintenance work to the required report.

Parameters: array pointer and number of records.

Returns: No value; writes a text file.

8. DeleteServiceArray Function

void DeleteServiceArray(ServiceTask **, int *)

Purpose: Deallocates the dynamic task array and resets pointer and size.

Parameters: array pointer and size pointer.

Returns: No value.

PRINT SCREENS

Main menu:

```
Wind Turbine Maintenance Planner
ES216BB
Evaluation 1

1. Load service tasks
2. Display one task
3. Display due tasks
4. Export maintenance plan
5. Exit
Choice:
```

No data loaded:

```
Wind Turbine Maintenance Planner
ES216BB
Evaluation 1

1. Load service tasks
2. Display one task
3. Display due tasks
4. Export maintenance plan
5. Exit
Choice: 2

No service tasks are loaded.

Press any key to continue...
```

Load service tasks:

```
Wind Turbine Maintenance Planner
ES216BB
Evaluation 1

1. Load service tasks
2. Display one task
3. Display due tasks
4. Export maintenance plan
5. Exit
Choice: 1

Input file: turbine_service.txt
5 service tasks loaded.

Press any key to continue...
```

Display service priority:

```
Wind Turbine Maintenance Planner
ES216BB
Evaluation 1

1. Load service tasks
2. Display one task
3. Display due tasks
4. Export maintenance plan
5. Exit
Choice: 2

Turbine 201: SCHEDULED, 65 days since service

Press any key to continue...
```

Display due maintenance:

```
Wind Turbine Maintenance Planner
ES216BB
Evaluation 1

1. Load service tasks
2. Display one task
3. Display due tasks
4. Export maintenance plan
5. Exit
Choice: 3

Tasks due at 180 days or earlier:
Turbine 202: DUE, 210 days since service
Turbine 203: CRITICAL, 10 days since service
Turbine 204: CRITICAL, 360 days since service
Turbine 205: CRITICAL, 90 days since service

Press any key to continue...
```

Export maintenance plan:

```
Wind Turbine Maintenance Planner
ES216BB
Evaluation 1

1. Load service tasks
2. Display one task
3. Display due tasks
4. Export maintenance plan
5. Exit
Choice: 4

Maintenance plan written to turbine_maintenance_plan.txt

Press any key to continue...
```

Exit application:

```
Wind Turbine Maintenance Planner
ES216BB
Evaluation 1

1. Load service tasks
2. Display one task
3. Display due tasks
4. Export maintenance plan
5. Exit
Choice: 5

Maintenance planner closed.

Press any key to continue...
```

Text File Input:

turbine_service.txt - Notepad

```
201 65 430 R 6.5
202 210 920 R 12.0
203 10 1050 F 42
204 360 760 R 18.0
205 90 500 F 13
```

Ln 1, Col 1 | 100% | Unix (LF) | UTF-8

Text File Output:

turbine_maintenance_plan.txt - Notepad

```
WIND TURBINE MAINTENANCE PLAN
202 DUE
203 CRITICAL
204 CRITICAL
205 CRITICAL
```

Ln 1, Col 1 | 100% | Windows (CRLF) | UTF-8

ANNEXURE A – MARK ALLOCATION

Note: Score range is 0 - 4: 0-none, 1-poor, 2-average, 3-good, 4-excellent

TEST RUBRIC	SCORE [0-4]	WEIGHT [%]
C++ CODE EVALUATION		**50**
0. Structures, types, static and dynamic access		**8**
1. Function1		**6**
2. Function2		**4**
3. Function3		**6**
4. Function4		**5**
5. Function5		**5**
6. Function6		**4**
7. Function7		**3**
8. Function8		**4**
9. Compile and runtime stability		**5**
TOTAL		**50**

Graduate Attribute	GA Number	GA Score [0-5]
Application of scientific and engineering knowledge	GA2	0,1,4
Engineering methods, skills, tools, including information technology	GA5	1,2,3,5,6
Impact of Engineering Activity	GA7	4,5,6
Engineering Professionalism	GA10	7,8,9

ANNEXURE B – INFORMATION SHEET

Data types: void, char, short, int, float, double

Data Type modifiers: const, auto, static, unsigned, signed

Arithmetic operators: * / % + -

Relational operators: < <= > >= == !=

Assignment operator: = += -= *= /= %= &= ^= |= <<= >>=

Logic operators: && || !

Bitwise logic operators: & | ^ ~ << >>

Pointer operators: Derefernce: * Address: &

Control Structures:

- **IF** Selection: if (condition) { … };
- **IF ELSE** Selection: if (condition) { … } else { … };
- **WHILE** Loop: while (condition) { … };
- **DO WHILE** loop: do { … } while (condition);
- **FOR** Loop: for (initial value of control variable; loop condition; increment of control variable) { … }
- **SWITCH** Selection: switch (control variable){ case ‘value’: … ; break; default: … ; break; }

Functions: return_data_type function_name (parameters) { … };

Arrays:

- One dimensional: data_type variable_name[size];
- Two dimensional: data_type variable_name [x_size][y_size];

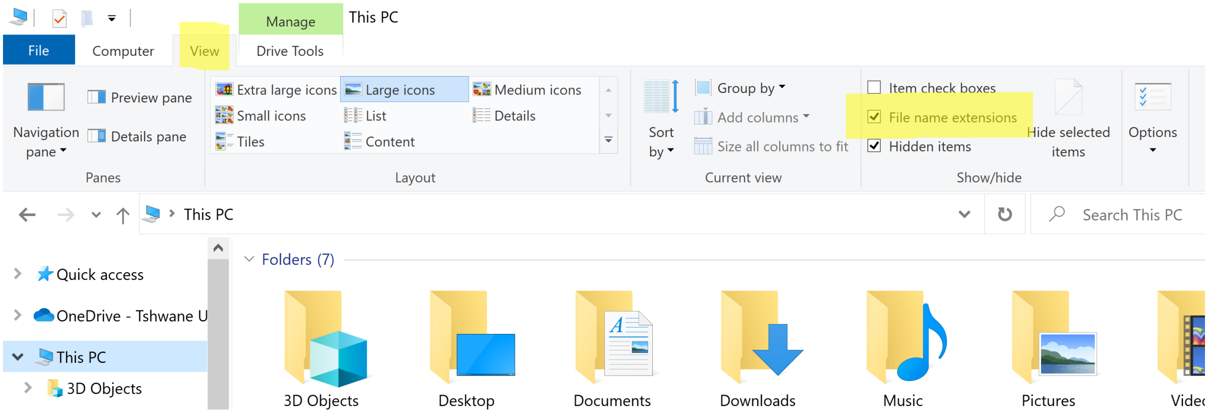
ANNEXURE C – ASCII TABLE

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040		Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174			|
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177	&#127;	DEL

Source: www.LookupTables.com

ANNEXURE D – HOW TO RUN THE SHOWCASE FILE

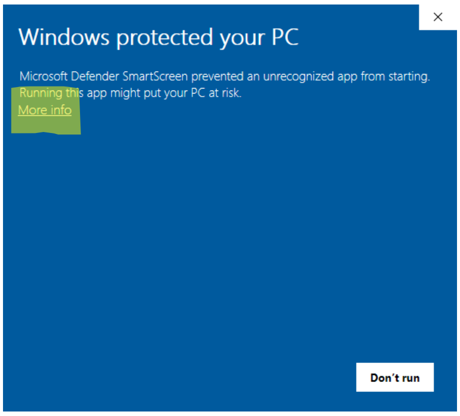
1. Enable file extensions (see highlighted in yellow)



2. Change the name from **"Showcase.old"** to **"Showcase.exe"**

3. Run the **"ShowcaseEV.exe"** by double-clicking on the icon.

4. Windows may show the following. Click on **"More info"**



5. Click on **"Run anyway"**